MACHINE LEARNING · ENSEMBLE METHODS

Gradient Boosting.

A complete, from-scratch tutorial — the intuition, the math, the code, and the pro-level hyperparameter tuning playbook. Built for readers who want to actually *understand* the algorithm, not just call `.fit()`.

BASIC INTERMEDIATE ADVANCED PRO

Regression · Classification · Hyperparameter Tuning · scikit-learn

Dark-mode edition

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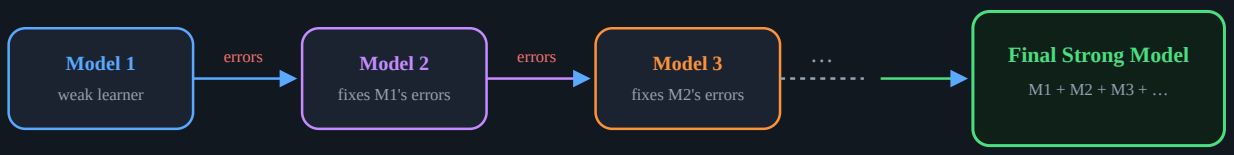
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01 What Is Boosting?

BASIC

Boosting is a machine learning strategy where you build one big, strong model out of many small, weak models — but you don't build them all at once, and you don't build them independently. You add them one at a time, **in sequence**, and each new model is trained specifically to fix the mistakes of the ones that came before it. This is called **sequential stage-wise addition**.



Each model is trained AFTER seeing what the previous ones got wrong — that's the essence of boosting.

Fig 1.1 — Sequential stage-wise addition: small models chained together into one strong model.

You may already be familiar with **AdaBoost**, another popular boosting algorithm. Both AdaBoost and Gradient Boosting follow this same sequential philosophy, but they disagree on *how* to tell the next model about the previous model's mistakes:

AdaBoost's approach

Rows the model got wrong are given a **higher weight**. The next weak learner is forced to pay more attention to those high-weight rows.

Gradient Boosting's approach

Instead of re-weighting rows, it directly computes the numeric **error (residual)** for every row, and trains the next model to *predict that error*.

🔑 KEY IDEA

Gradient Boosting doesn't just say "pay attention to this row" — it says "here is exactly *how wrong* we were on this row, now go predict that gap." This is a more direct, more mathematical way of passing on mistakes, and it's why Gradient Boosting typically performs even better than AdaBoost. An optimized, production-grade version of this idea is called **XGBoost**, which today is one of the most competitive algorithms in applied machine learning.

Gradient Boosting works for both **regression** and **classification**. We'll build the intuition using regression first, because the math is simpler and every idea transfers directly to classification later (Section 12).

02 The Core Idea, In One Picture

BASIC

Before touching any data, let's fix the mental model. Every Gradient Boosting model is built from three repeating ingredients:

① A starting guess

A very simple first model — for regression, literally just the average of the target column.

② Measure the error

Compute `Actual − Predicted` for every row. This is called a **pseudo-residual**.

③ Predict the error

Train a small Decision Tree whose job is not to predict the target — but to predict *how wrong* the current model is.

Then you repeat ② and ③ over and over, each time adding the new tree's (shrunk) prediction on top of everything built so far. The residual keeps shrinking toward zero, which means your combined prediction keeps getting closer to the truth.



Fig 2.1 — The Gradient Boosting loop: predict → measure residual → fit a tree to the residual → add it back (scaled) → repeat.

✅ PLAIN-ENGLISH SUMMARY

"Start with a dumb guess. Look at how wrong you were. Train a tree to guess *the wrongness*. Nudge your prediction using that tree's guess. Repeat until the wrongness is tiny."

03 The Worked Example — Predicting Salary

BASIC

We'll use one small toy dataset for the entire hand-worked walkthrough (Sections 04–09), exactly the way you'd work it out on a whiteboard. Two inputs — **CGPA** and **IQ** — predict one output: **LPA** (salary, "Lakhs Per Annum").

STUDENT	CGPA	IQ	LPA (ACTUAL SALARY)
S1	8.1	90	3.6
S2	6.5	100	3.2
S3	7.5	110	8.0
S4	9.0	108	6.0
S5	5.8	80	3.2

We'll build a Gradient Boosting model made of **3 small models** chained together (in real projects you'd typically use 100+, but three is enough to see every moving part clearly).

> 🔑 **THE ONE RULE TO REMEMBER**
>
> For a **regression** problem, Model 1 in Gradient Boosting is *always* just the mean of the output column. It isn't a Decision Tree, it isn't AdaBoost — it's a single number, sometimes called a "leaf" model, that predicts the same value for every single row regardless of input.

04 Model 1 — The Mean Baseline

BASIC

Our first model takes no input at all — it just always outputs the average LPA:

$$\text{mean(LPA)} = (3.6 + 3.2 + 8.0 + 6.0 + 3.2) / 5 = \mathbf{4.8}$$

This single number, 4.8, is the entire "Model 1." Every student — no matter their CGPA or IQ — gets a predicted salary of 4.8 LPA from this model.

STUDENT	ACTUAL LPA	PREDICTION 1
S1	3.6	4.8
S2	3.2	4.8
S3	8.0	4.8
S4	6.0	4.8
S5	3.2	4.8

📐 **WHY THE MEAN? (PREVIEW)**

The mean is the value that *minimizes squared error* across a set of numbers. Gradient Boosting's first model is chosen to minimize the loss function before any features are even used — for the standard squared-error loss, that minimizer is exactly the mean. We'll dig into this properly in Section 05.

05 Pseudo-Residuals & the Loss Function

INTERMEDIATE

Now we ask: how good is Model 1? To measure that, we need a **loss function** — a way of scoring how wrong a prediction is.

$$\text{Loss}(\text{Actual}, \text{Predicted}) = \text{Actual} - \text{Predicted} \quad \rightarrow \quad \text{called the } \textbf{Pseudo-Residual}$$

Contrast this with plain linear regression, which squares the error: (Actual − Predicted)². Gradient Boosting works with the raw difference at this stage.

STUDENT	ACTUAL	PREDICTION 1	RESIDUAL 1 = ACTUAL − PRED 1
S1	3.6	4.8	−1.2
S2	3.2	4.8	−1.6
S3	8.0	4.8	+3.2
S4	6.0	4.8	+1.2
S5	3.2	4.8	−1.6

These five numbers are the mistakes made by Model 1. Now comes the most important idea in the whole algorithm: how do we *hand off* these mistakes to the next model?

🔑 THE HAND-OFF TRICK

We don't tell Model 2 "predict the salary." We tell it: **"look at each student's CGPA and IQ, and predict how much error Model 1 made."** The output column for Model 2's training is not LPA — it's `Residual 1`. The better Model 2 predicts the error, the more we can correct our overall prediction.

Advanced note — why "pseudo"?

📐 MATH NOTE (ADVANCED / PRO)

Gradient Boosting is really **gradient descent in function space**. For squared-error loss $L(y, F) = \frac{1}{2}(y - F)^2$, the negative gradient with respect to the prediction F is:

$$-\partial L/\partial F = (y - F)$$

That negative gradient is exactly the residual we computed above. So "pseudo-residual" really means *"the negative gradient of the loss, which for squared error happens to equal the plain residual."* This is why the algorithm generalizes to classification and other loss functions (Section 12) — you always fit the next tree to the negative gradient, whatever the loss function is.

06 Model 2 — Fitting a Tree to the Errors

INTERMEDIATE

Model 2 is a small **Decision Tree Regressor**. Its inputs are CGPA and IQ — exactly the same features as before — but its **target is Residual 1**, not LPA.

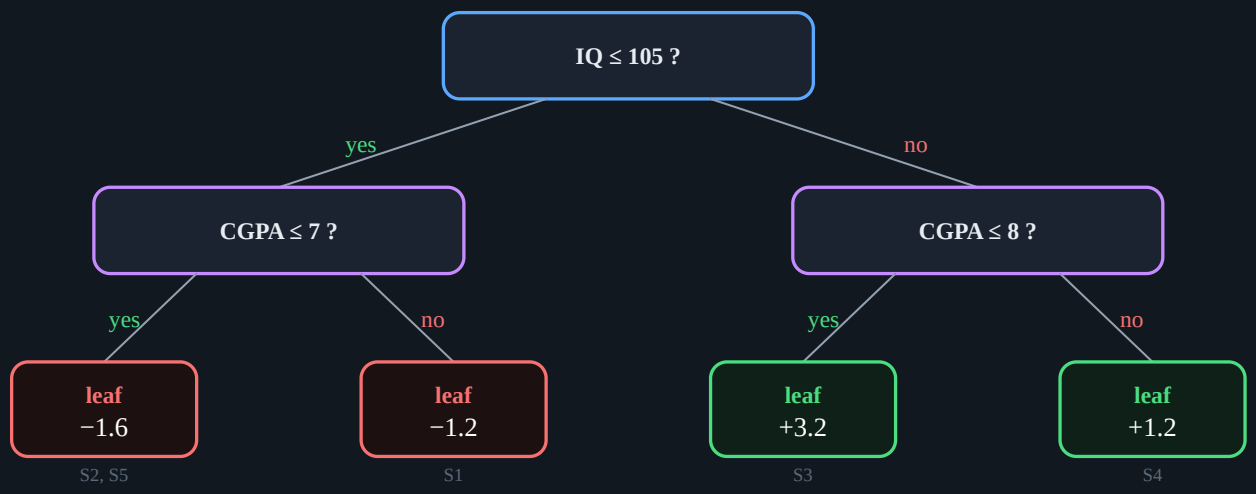


Fig 6.1 — Model 2, a Decision Tree Regressor trained with target = Residual 1. It splits on IQ first, then CGPA.

Sending every student down this tree gives us **Prediction 2** — and because our toy dataset is so small, the tree memorizes the residuals almost perfectly:

STUDENT	CGPA	IQ	RESIDUAL 1	PREDICTION 2 (TREE OUTPUT)
S1	8.1	90	−1.2	−1.2
S2	6.5	100	−1.6	−1.6
S3	7.5	110	+3.2	+3.2
S4	9.0	108	+1.2	+1.2
S5	5.8	80	−1.6	−1.6

💡 **WHY TREES, AND WHY NOT DECISION STUMPS?**

Research on boosting has shown Gradient Boosting performs best with trees that have between **8 and 32 leaf nodes** — not the single-split "decision stumps" (depth-1 trees) that AdaBoost uses. Small datasets favor the lower end (≈8), large datasets can push toward 32. We're only using 4 leaves here purely because our toy dataset has 5 rows — in real code you'd set `max_leaf_nodes=8` or tune `max_depth`.

07 Combining Models & the Overfitting Trap

INTERMEDIATE

With two models built, how do we make an actual salary prediction? Simply **add the outputs together**:

$$\text{Prediction} = \text{Model 1 output} + \text{Model 2 output}$$

STUDENT	MODEL 1	MODEL 2	COMBINED PREDICTION	ACTUAL
S1	4.8	−1.2	3.6	3.6
S2	4.8	−1.6	3.2	3.2

Look at that — the combined prediction matches the actual salary **exactly**. That might look like a win... but it's actually a red flag.

> ⚠️ **THE OVERFITTING TRAP**
>
> Since our second tree simply memorized the residuals of a tiny 5-row dataset, the combined model reproduces the training data perfectly. That's **overfitting**: the model has learned the training rows by heart, not the underlying pattern. On a new student it has never seen, this model will likely make poor predictions, because it never learned to *generalize* — only to *memorize*.

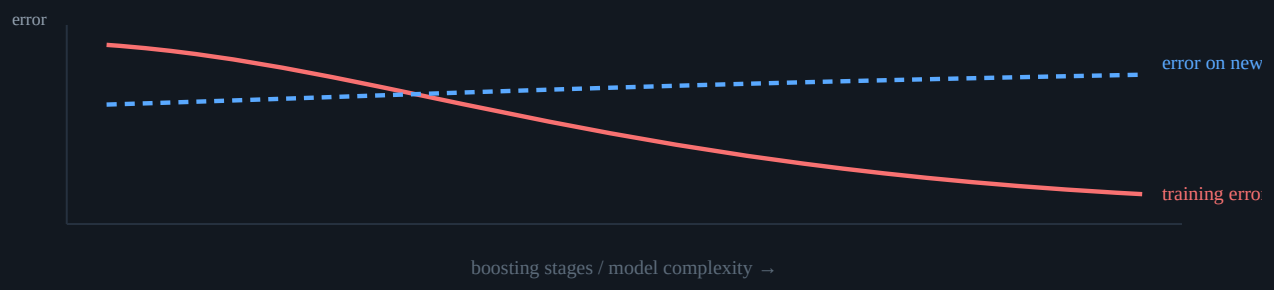


Fig 7.1 — Without control, training error keeps shrinking toward zero while error on unseen data can start rising: classic overfitting.

So how do we get the benefit of boosting without memorizing the training set? The fix is delightfully simple: **take smaller steps.**

08 The Learning Rate Fix

INTERMEDIATE

Instead of adding a new tree's full output, we **shrink** it by multiplying with a small constant called the **learning rate** (η), typically between 0 and 1 — commonly starting around **0.1**.

$$\text{Prediction} = \text{Model 1} + \eta \times \text{Model 2}$$

η (eta) is the learning rate. With η = 0.1, only 10% of each new tree's correction is actually applied.

Let's redo Student 1's prediction with η = 0.1:

$$4.8 + 0.1 \times (-1.2) = 4.8 - 0.12 = \mathbf{4.68}$$

Compare to the actual salary of 3.6 — no longer a perfect match, and that's exactly the point.

> 🔑 **WHY "IMPERFECT" IS GOOD HERE**
>
> By deliberately under-correcting at every stage, we force the model to take many small, cautious steps toward the answer rather than one giant memorizing leap. Each step generalizes a little better. This is the single biggest lever against overfitting in Gradient Boosting — smaller learning rate = more cautious, usually more generalizable, but needs more trees (`n_estimators`) to converge.



Fig 8.1 — A large learning rate reaches the training target in one jump (overfits); a small learning rate creeps toward it gradually.

Watch the residual shrink

After adding Model 2 with η = 0.1, we recompute the residual — call it Residual 2 — using the *combined* prediction so far:

$$\text{Residual 2} = \text{Actual} - (\text{Model 1} + \eta \times \text{Model 2})$$

STUDENT	ACTUAL	COMBINED PRED (H=0.1)	RESIDUAL 2
S1	3.6	4.68	−1.08
S2	3.2	4.64	−1.44
S3	8.0	5.12	+2.88
S4	6.0	4.92	+1.08
S5	3.2	4.64	−1.44

Notice the pattern: −1.2 became −1.08, −1.6 became −1.44, +3.2 became +2.88 — every residual moved *toward zero*. That's exactly what we want: each stage nudges us closer to the truth without leaping all the way there.

09 Model 3 and the General Algorithm

ADVANCED

Model 3 repeats the exact same recipe as Model 2 — a fresh Decision Tree, same inputs (CGPA, IQ), but this time trained to predict **Residual 2**:

STUDENT	RESIDUAL 2 (TARGET)	PREDICTION 3 (TREE OUTPUT)
S1	−1.08	−1.08
S2 / S5	−1.44	−1.44
S3	+2.88	+2.88
S4	+1.08	+1.08

With three models built, the prediction formula extends naturally:

$$\text{Prediction} = \text{Model 1} + \eta\times\text{Model 2} + \eta\times\text{Model 3}\ (\ +\ \eta\times\text{Model 4} + \ldots \text{ if you kept going}\ )$$

The learning rate η is the *same* for every model from Model 2 onward — this is a key difference from AdaBoost, where every weak learner gets its own, different weight (see Section 11).

Predicting for a brand-new student

Take a student the model has never seen: CGPA = 5.0, IQ = 60.

1. Model 1 → **4.8** (always, regardless of input)
2. Model 2 tree: IQ 60 ≤ 105 → yes; CGPA 5.0 ≤ 7 → yes → leaf = **−1.6**
3. Model 3 tree: same path → leaf = **−1.44**

$$4.8 + 0.1\times(-1.6) + 0.1\times(-1.44) = 4.8 - 0.16 - 0.144 = \approx \mathbf{4.50\ LPA}$$

📐 THE GENERAL ALGORITHM (PRO-LEVEL FORMALISM)

Generalizing everything above, for M boosting stages, Gradient Boosting for regression is:

$$F_0(x) = \text{mean}(y)$$
$$\text{for } m = 1 \text{ to } M:$$
$$r_i = y_i - F_{m-1}(x_i) \quad \text{(pseudo-residuals)}$$
$$h_m(x) = \text{fit a tree to } (x_i, r_i)$$
$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$
$$\text{Final model: } F_M(x)$$

This is literally functional gradient descent: at every step you move the current function F a little bit in the direction that most reduces the loss, where "direction" is represented by a freshly-trained tree instead of a plain numeric gradient vector.

✅ SANITY CHECK AS M GROWS

As you add more and more trees (M ↑), the pseudo-residuals keep shrinking toward zero and $F_M(x)$ keeps approaching the true target — but push M too high with too large a learning rate, and you're back in overfitting territory. Balancing `n_estimators` against `learning_rate` is the central tension of Gradient Boosting, and it's the whole subject of Part IV.

10 Geometric Intuition & a Minimal Python Walkthrough

ADVANCED

Zooming out from the toy 5-row example: imagine a real, noisy, non-linear dataset with 100 points. Here's what happens visually as more boosting stages are added.

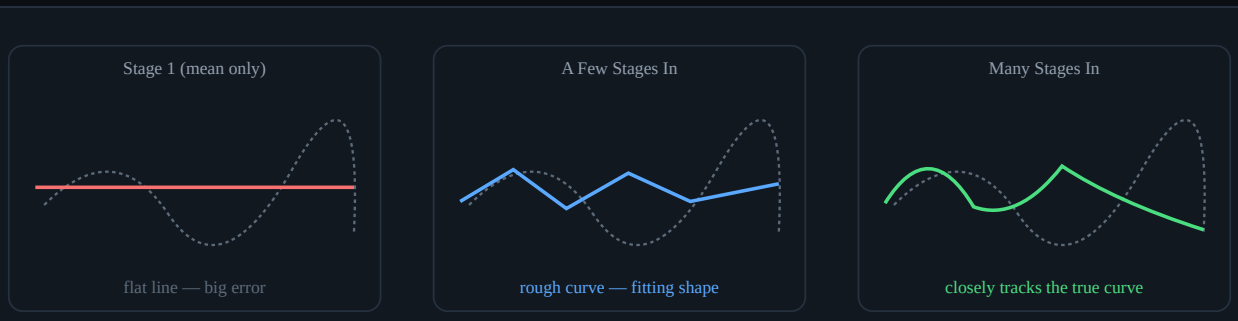


Fig 10.1 — Illustrative progression: with each boosting stage, the cumulative prediction (solid line) bends closer to the true underlying pattern (dashed).

Left unchecked and run for too many stages with too high a learning rate, this same curve eventually starts wiggling through every noisy data point — the overfitting trap from Section 07, now visible geometrically.

Minimal from-scratch regression code

Here's the entire algorithm from Section 09's pseudocode, translated into runnable Python with scikit-learn's `DecisionTreeRegressor` as the weak learner:

```python
import numpy as np
from sklearn.tree import DecisionTreeRegressor

def gradient_boost_fit(X, y, n_estimators=50, learning_rate=0.1, max_leaf_nodes=8):
    models = []
    F0 = np.mean(y)                      # Model 1: the mean
    models.append(F0)
    current_pred = np.full(y.shape, F0)

    for m in range(n_estimators):
        residual = y - current_pred        # pseudo-residuals
        tree = DecisionTreeRegressor(max_leaf_nodes=max_leaf_nodes)
        tree.fit(X, residual)              # fit tree TO THE RESIDUAL
        models.append(tree)
        current_pred += learning_rate * tree.predict(X)   # shrink & add

    return models

def gradient_boost_predict(models, X, learning_rate=0.1):
    F0, trees = models[0], models[1:]
    pred = np.full(X.shape[0], F0)
    for tree in trees:
        pred += learning_rate * tree.predict(X)
    return pred
```

💡 **WHAT YOU'D SEE RUNNING THIS ON REAL DATA**

Plot the fitted curve after 1, 5, 20, and 100 boosting stages: the first plot is a flat line (just the mean), the middle plots progressively hug the data's shape, and by 100 stages with a learning rate that's too high you'll typically see the curve start "chasing" individual noisy points — the visual signature of overfitting.

11 AdaBoost vs. Gradient Boosting

INTERMEDIATE

Both are boosting algorithms built from sequential weak learners, but they differ in two fundamental ways:

ASPECT	ADABOOST	GRADIENT BOOSTING
Weak learner shape	Decision **stumps** (max depth = 1)	Small trees with **8–32 leaf nodes**
How mistakes are passed on	Misclassified rows get **higher sample weight**	Next model directly fits the **numeric residual**
Model importance	Each weak learner gets its **own weight** (W1, W2, W3…) based on its accuracy	**Same learning rate** applied uniformly to every model
What's minimized	Exponential loss (implicitly)	Any differentiable loss function (explicitly, via gradients)
Sensitivity	Very sensitive to noisy data / outliers	More robust, but slower to train

⚠️ COMMON MIX-UP

People sometimes assume Gradient Boosting re-weights rows like AdaBoost does. It doesn't. There is no per-row weighting step in standard Gradient Boosting — errors are passed on purely through the numeric target (the residual) that the next tree is trained on.

12 From Regression to Classification

ADVANCED

Everything above was built for regression, where squared error made the residual trick clean: negative gradient = actual − predicted. For **classification**, the same skeleton (start simple → compute negative gradient → fit a tree to it → shrink & add → repeat) still applies, but two things change:

1. The starting model

Instead of the mean of y, Model 1 starts from the **log-odds** of the class distribution — the classification equivalent of "the best constant guess."

2. The loss function

Squared error is replaced by **log loss / deviance**. Each tree still fits the negative gradient of this loss — it's just a differently-shaped residual.

The predictions from all the trees are summed exactly as before, then passed through a **sigmoid** (binary classification) or **softmax** (multi-class) function to turn the raw sum into a probability.

🔑 THE REASSURING TAKEAWAY

You do not need to learn a second algorithm for classification. It is literally the same loop from Section 09's pseudocode, with a different loss function plugged in. This is exactly why Gradient Boosting frameworks (scikit-learn, XGBoost, LightGBM) expose both a `Regressor` and a `Classifier` class that share almost all their hyperparameters.

13 Hands-On: GradientBoostingClassifier in scikit-learn

ADVANCED

Let's put it into practice on a real dataset — the built-in **Wine** dataset from scikit-learn (3 wine classes, 13 chemical features).

```
from sklearn import datasets
from sklearn.model_selection import train_test_split, cross_val_score, GridSearchCV
from sklearn.ensemble import GradientBoostingClassifier

wine = datasets.load_wine()
X, y = wine.data, wine.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=17
)

gbc = GradientBoostingClassifier()
gbc.fit(X_train, y_train)

# Baseline performance with default hyperparameters
baseline = cross_val_score(gbc, X_train, y_train, cv=3, n_jobs=-1).mean()
print(baseline)   # e.g. ~0.922
```

A baseline of ~0.92 accuracy is solid, but almost never optimal — that's where hyperparameter tuning comes in:

```
param_grid = {
    'n_estimators':   [10, 50, 100, 500],
    'learning_rate':  [0.001, 0.01, 0.1, 1.0],
    'max_depth':      [3, 5, 7, 9],
}

search = GridSearchCV(gbc, param_grid, cv=3, n_jobs=-1)
search.fit(X_train, y_train)

print(search.best_params_)   # e.g. {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 500}
print(search.best_score_)    # e.g. ~0.943 — an improvement over the baseline
```

> ⚠️ **WATCH THE SEARCH SIZE**
>
> 4 values × 4 values × 4 values = 64 combinations, multiplied by `cv=3` folds = **192 model fits** for this one grid. GridSearchCV tests every combination exhaustively, so it's the slowest tuning strategy — great for small grids, painful for large ones. Section 17 covers faster alternatives.

14 The Hyperparameters, One by One

PRO

Gradient Boosting's real power only shows up once you know what each knob does. Here is every major hyperparameter, what it controls, and its default.

HYPERPARAMETER	WHAT IT CONTROLS	DEFAULT
`n_estimators`	Number of boosting stages (trees) to add. More = better fit, but higher overfitting risk.	100
`learning_rate`	Shrinkage applied to every tree's contribution (aka η / eta).	0.1
`max_depth`	Maximum depth of each individual tree. Shallow → underfit; deep → overfit.	3 (sklearn classifier)
`min_samples_split`	Minimum samples required before a node is allowed to split.	2
`min_samples_leaf`	Minimum samples required to form a leaf. Higher → smoother, less overfit trees.	1
`subsample`	Fraction of rows randomly sampled to train each tree (stochastic gradient boosting).	1.0
`max_features` / `colsample_bytree`	Fraction/number of features randomly considered per tree.	None (all)
`reg_alpha` / `reg_lambda`	L1 / L2 regularization on tree leaf weights (XGBoost / LightGBM).	0

🔑 **THE TWO THAT MATTER MOST**

`n_estimators` and `learning_rate` are joined at the hip: a smaller learning rate needs more estimators to reach the same fit, and vice versa. Everything else in the table above exists mainly to fight the overfitting that a high-capacity, high-`n_estimators` model risks.

15 The Professional Tuning Procedure

PRO

Kaggle grandmasters and production ML engineers rarely tune every hyperparameter at once with a single blind grid search. Instead, they follow a staged procedure:

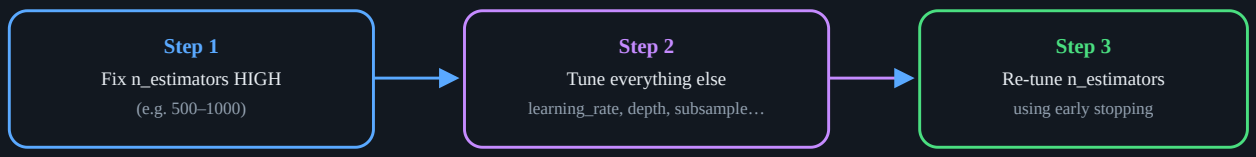


Fig 15.1 — The staged "pro" tuning procedure, instead of tuning everything simultaneously in one huge grid.

1. **Fix** `n_estimators` **high** (500–1000) as a placeholder so the model has enough capacity to express whatever the other hyperparameters allow.
2. **Tune the rest** — `learning_rate`, `max_depth`, `subsample`, `min_samples_split`, etc. — with that high estimator count held constant.
3. **Re-tune the number of trees** using **early stopping**: keep adding trees only while the validation score keeps improving, and stop once it hasn't improved for, say, 15–20 rounds.

> 💡 **WHY THIS ORDER WORKS**
>
> Tuning `n_estimators` jointly with everything else wastes enormous compute, because the "right" number of trees depends heavily on the learning rate you land on. Decoupling the two — find good shape parameters first, then let early stopping decide exactly how many trees you need — is dramatically more efficient.

16 Search Spaces & Regularization Cheat Sheet

PRO

HYPERPARAMETER	RECOMMENDED SEARCH SPACE	NOTES
`learning_rate`	0.001 – 0.05 **(log scale)**	0.01 is a strong default starting point. Don't search too low without raising `n_estimators` to compensate.
`max_depth`	1 – 3 (small data) · 1 – 6 (large data)	Gradient Boosting thrives with shallow, "stump-like" trees — unlike Random Forests, which prefer deep trees.
`subsample`	0.5 – 1.0	0.7 is a highly reliable default; don't go too low or trees see too little data.

📐 ALWAYS SEARCH LEARNING_RATE ON A LOG SCALE

The effect of moving learning_rate from 0.001 → 0.01 is much bigger than moving it from 0.5 → 0.51 — a linear grid wastes most of its samples where they matter least. A log-uniform search (e.g. `trial.suggest_float(..., log=True)` in Optuna) samples proportionally more low values, matching how sensitive the model actually is.

Regularization: L1 vs L2

L2 (`reg_lambda`) — default choice

Use when you expect high collinearity among features but want to keep all of them. Smoothly shrinks leaf weights.

L1 (`reg_alpha`)

Use when your dataset has a lot of noisy, irrelevant features — L1 pushes some leaf weights toward zero, effectively selecting a subset.

Rule of thumb: don't use both at full strength simultaneously — pick the one that matches your data's problem (collinearity → L2, noisy/irrelevant features → L1).

Cross-validation discipline

⚠️ ALWAYS USE K-FOLD CROSS-VALIDATION

Gradient Boosting's hyperparameter space is large enough that a single train/validation split can easily give you a lucky (or unlucky) score. Use k-fold CV for anything you're about to trust. **Exception:** for time-series data, never shuffle — use a nested or sliding-window CV scheme instead, so you never train on the future to predict the past.

17 GridSearch vs. RandomSearch vs. Optuna

PRO

STRATEGY	HOW IT WORKS	BEST FOR
GridSearchCV	Exhaustively tries every combination in a fixed grid.	Small grids where you can afford to test everything.
RandomizedSearchCV	Randomly samples a fixed number of combinations from given distributions.	Larger spaces where exhaustive search is too slow.
Optuna (Bayesian)	Learns from previous trials to intelligently propose the next set of hyperparameters.	Large, expensive search spaces — converges to a good result in far fewer trials.

> 💡 **THE PROFESSIONAL'S CHOICE**
>
> Bayesian optimization frameworks like **Optuna** or **Hyperopt** are generally preferred over grid/random search for Gradient Boosting because the search space is large and expensive to evaluate — Bayesian methods use the results of earlier trials to intelligently decide what to try next, often reaching a strong result in ~100 trials that grid search would need thousands of fits to match.

```python
import optuna
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.model_selection import cross_val_score

def objective(trial):
    param = {
        'n_estimators': 500,  # fixed high; early stopping decides the real count later
        'learning_rate': trial.suggest_float('learning_rate', 0.001, 0.05, log=True),
        'max_depth': trial.suggest_int('max_depth', 1, 6),
        'subsample': trial.suggest_float('subsample', 0.5, 1.0),
        'min_samples_split': trial.suggest_int('min_samples_split', 2, 10),
    }
    clf = GradientBoostingClassifier(**param, random_state=42)
    return cross_val_score(clf, X_train, y_train, cv=5, scoring='accuracy').mean()

study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=50)

print("Best Parameters:", study.best_params)
# Finally: train with best params + early_stopping_rounds=20 to lock in n_estimators
```

18 Final Cheat Sheet & Glossary

PRO

One-page mental model

1. Model 1 = mean of y (regression) or log-odds (classification).
2. Residual = actual − predicted (or, more generally, the negative gradient of the loss).
3. Fit a small tree (8–32 leaves) to predict that residual.
4. Add the tree's prediction back, scaled by the learning rate.
5. Repeat for `n_estimators` rounds, or until early stopping says stop.
6. Tune: fix trees high → tune shape params → re-tune trees with early stopping.

Glossary

TERM	MEANING
Pseudo-residual	Actual − Predicted; more generally, the negative gradient of the loss function w.r.t. the current prediction.
Weak learner	A small, individually mediocre model (e.g. a shallow decision tree) combined with others to form a strong model.
Learning rate (η)	Shrinkage factor applied to every new tree's output before adding it to the running prediction.
Stage-wise addition	Building the ensemble one model at a time, each depending on the state left by the previous ones.
Early stopping	Halting training once validation performance stops improving for a set number of rounds.
Shrinkage	Another name for the learning rate's effect — "shrinking" each tree's contribution.
Deviance / log loss	The loss function used for classification variants of Gradient Boosting, analogous to squared error for regression.

✅ **YOU'RE READY FOR**

XGBoost, LightGBM, and CatBoost — all three are optimized, production-grade implementations of exactly the algorithm you just learned, with extra tricks (regularization, histogram-based splitting, native categorical support) layered on top of the same core loop: predict → residual → fit a tree → shrink → add → repeat.